

Readable SQL and Reusable Results

Name each reasoning step, then choose how long its result should live.

CTEs and subqueries

BigQuery sessions

SQLite and Snowflake objects

Readable SQL exposes the reasoning

One dense block

Filters, calculations, and business rules are mixed together.
A reviewer must reconstruct the logic.

same result

Named steps

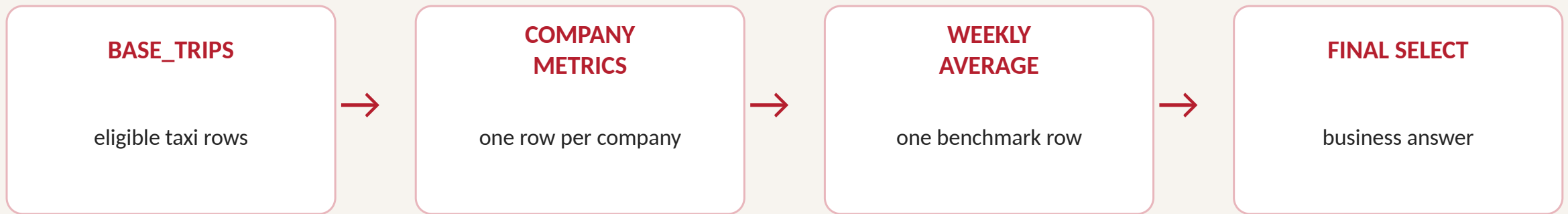
Filter the week. Build metrics. Compare with a benchmark.
Present the final grain.

The value is not fewer lines. The value is visible reasoning.

Today you will choose the right SQL scope

- 1 Name and test query steps with CTEs.
- 2 State the output grain before interpreting.
- 3 Use BigQuery sessions for temporary tables.
- 4 Distinguish CTEs, temp tables, views, and CTAS.
- 5 Transfer DDL and query reasoning to Snowflake.

A CTE pipeline has a beginning, middle, and end



Each named step has a grain you can test independently.

Grain tells you what one result row means

Source grain

One row per taxi trip

Metric grain

One row per company

Final grain

One row per qualifying company

If you cannot finish the sentence “one row represents...”, do not publish the result.

BigQuery sessions retain temp tables, not DML writes

Session supports

SELECT and WITH CTEs
Temp tables across query entries
@@session_id
Public dataset analysis

Sandbox limits

INSERT, UPDATE, DELETE, MERGE
Streaming ingestion
Persistent DML workflows

Keep the query tab open. Use SQLite or your Snowflake schema for DDL and inserts.

One source table means no join is required

Question

Which companies have the most trips
in the selected week?

Available fields

company
trip_start_timestamp
trip_total

Decision

Filter, group, order.
No join needed.

Join only when the question needs a field that the source does not contain.

Cost-aware SQL follows a repeatable loop

- 1 Confirm the fully qualified source table.
- 2 Select only the columns needed by the question.
- 3 Use the verified 2023 date window.
- 4 Inspect the byte estimate before running.
- 5 Validate row count, grain, and business meaning.

CTAS stores rows; a view stores a query

CTAS snapshot

CREATE TABLE ... AS SELECT

Stores the rows returned at creation time.

View

CREATE VIEW ... AS SELECT

Reruns the saved query against current source rows.

CTAS stores a snapshot. A view stays current. A temp table ends with the session.

The logic transfers, while object choices differ

SQLite

CREATE TABLE and INSERT
CTAS and views
Local database file

BigQuery

CTEs and sessions
Logical and materialized views
Managed and external tables

Snowflake

Permanent, transient, temporary
Standard and secure views
Personal schema

Choose one statement, one session, or a persistent object based on reuse and freshness.

Close every query with five explanations

- 1 The business question I answered.
- 2 The source table and selected date window.
- 3 What one output row represents.
- 4 The validation and byte estimate I checked.
- 5 One risk and the next check before action.

Today: create in SQLite, repeat in Snowflake, then prepare for COPY INTO.